SECTION BR

Brake System

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General Instructions

WARNING

Detailed information on standard workshop and safety procedures, and general servicing operations is not included in this manual, which has been prepared to assist qualified service personnel. ARGO assumes no responsibility or liability for PERSONAL INJURY or VEHICLE DAMAGE which results from any servicing procedure performed, including those instructions outlined in this manual. Before performing a servicing operation, an individual must have determined to his/her satisfaction that a personal injury or vehicle damage will not result from the servicing procedure or tools selected.

WARNING

Always refer to the correct PARTS MANUAL in accordance with the specific model and serial number indicated on the serial tag plate of your vehicle. This is essential to locating and ordering the correct components that were used during that particular manufacturing period.

WARNING

All brake work should be performed by a qualified mechanic. Braking/Steering capability and vehicle performance will be seriously impaired if the braking system is not properly adjusted or maintained. Property damage and/or injury or death may result to the operator.

WARNING

Hydraulic fluid can cause serious irritation to skin and eyes. In case of contact, flush thoroughly with water and contact a doctor if eyes have been exposed.

CAUTION

Use only DOT 4 Brake Fluid in the hydraulic brake system. Use of other brake fluids may damage components and could void warranty on the brake system. Never allow dirt, water, used brake fluid or any other contaminants to enter the hydraulic brake system. Also, never combine different types of brake fluids keeping in mind they may be incompatible with each other.

CAUTION

Regular maintenance of the brake system is essential to both good vehicle performance and personal safety. Poorly maintained brakes can result in a decrease in braking/steering efficiency, vehicle performance or damaged brake discs.

Maintenance Schedule

Every 25 hours of vehicle operation Perform the servicing procedure**, Brake Pad Inspection.**

Every 50 hours of vehicle operation Perform the servicing procedure, **Inspect Brake Fluid Level/Condition.**

NOTE

The intervals shown in the Argo Maintenance Schedule Chart (found in the General Information section of this service manual) are based on average operating conditions. Vehicles which are subject to more severe use, wet or dusty conditions, will require more frequent servicing.

NOTE

Please refer to the appropriate illustrated parts manual for the vehicle being serviced. The parts manual is identified with either the vehicle's model number, serial number or both written on the front cover.

Hydraulic Brakes

General

Although the hydraulic brake system is self adjusting, the following require periodic attention:

Brake Fluid Level (Steering Brake Master Cylinders)

After every 50 hours of operation, check the brake fluid level by removing the master cylinder covers.

Brake Fluid Level (Service Brake Master Cylinders) Aurora Models Only

After every 50 hours of operation, check the brake fluid level by removing the master cylinder covers.

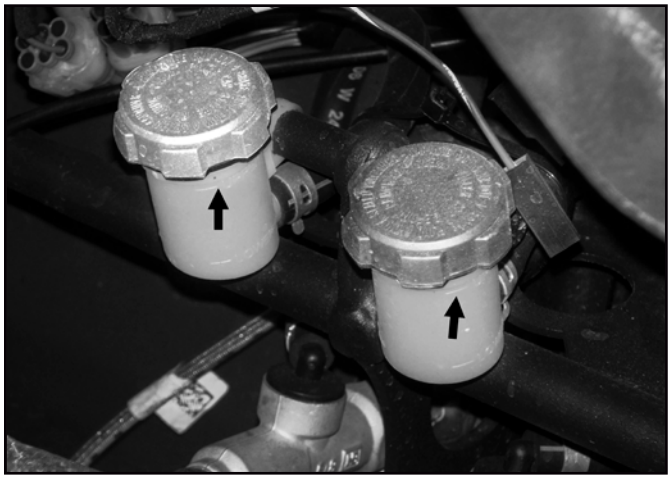
Checking The Hydraulic Brake Fluid Level (Steering Brakes), Aurora and Frontier 600 EU Models Only

IMPORTANT

Thoroughly clean the master cylinder cover and surrounding area before removal.

1. Locate the steering master cylinders in the engine compartment near the firewall at the steering column. When adding fluid, fill to mark on reservoir.

If the brake fluid is below this level:



2. Add only fresh clean Argo Brand Brake Fluid - DOT 4 (Argo Part No.130-108) to the correct level.
3. Replace the cover on each master cylinder, making sure the rubber gaskets are properly seated before tightening. Tighten snug by hand only.

CAUTION

Do not overfill the brake master cylinders. Overfilling can cause seal damage.

Hydraulic Brake Pad Inspection (Steering Brakes)

Inspect the steering brake pads after every 25 hours of operation. Worn, glazed or contaminated brake pads affect the efficiency of the brake system. To inspect the pads, first remove the firewall.

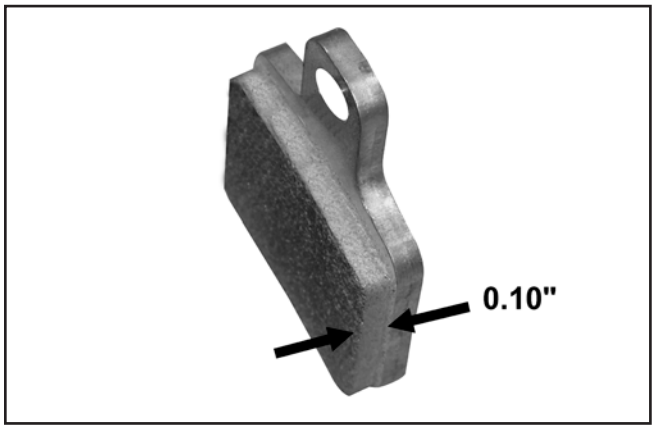
To remove the firewall:

1. Remove the six (6) fasteners securing the firewall using a socket and ratchet wrench.
2. Pull the firewall rear-ward moving the throttle cable clear of the area at the steering column that it is routed through.
3. Lift the firewall clear of the driving compartment.

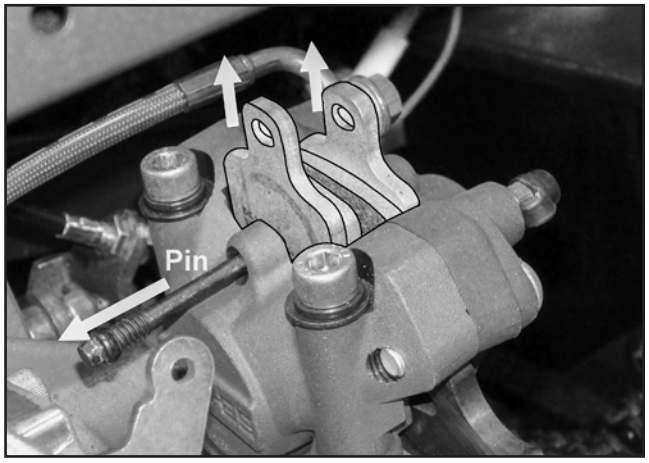
With the firewall removed, both hydraulic brake calipers are visible. Each caliper has 2 brake pads which are secured by a single bolt fastener. Inspect all 4 brake pads.

Replace the pads when:

- the brake lining material moulded to each metal backing plate is worn to 0.10" thickness.
- the pads are glazed and brake performance is affected.
- the pads are contaminated with lubricant, and brake performance is affected.



Replacing The Steering Brake Pads



Brake pads are easily replaced by removing the single bolt securing them within the brake calliper assembly and pulling each pad up and out of the calliper. Pistons have to be pushed back in first, to allow clearance for the new pads. Slip the new pads into the calliper and install the single bolt fastener to secure the pads in position. **Torque pad pin to 58.8-82.8 in-lb(6.6 Nm -9.5 Nm** Pump the steering handle bar a few times to the left and to the right to build up proper pressure and to locate the pads in the calliper assembly.

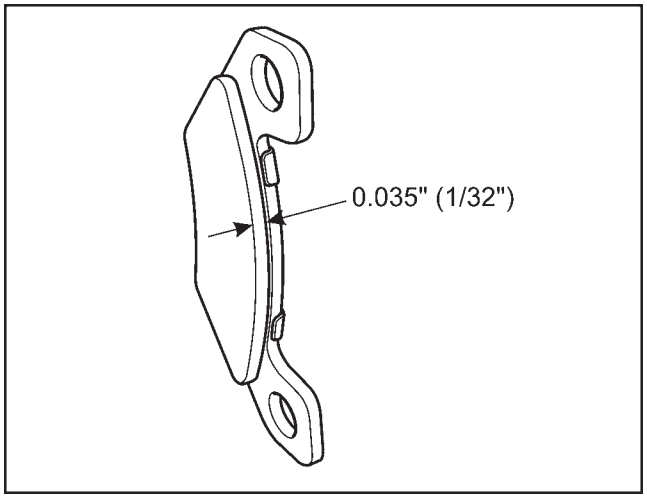
CAUTION

When replacing worn or contaminated pads, replace both pads in the brake caliper assembly even if both pads do not require replacing. Using one new and one old brake pad in a caliper assembly may provide poor braking/steering performance.

Brake Pad Inspection Procedure - (Service Brake Pads) Aurora Models Only

Inspect the brake pads after every 25 hours of operation. Worn, glazed or contaminated brake pads affect the efficiency of the brake system. To inspect the pads, first remove the firewall.

Each caliper has 2 brake pads which are secured by (2) 3/8” Socket Head bolts. Inspect all 4 brake pads.



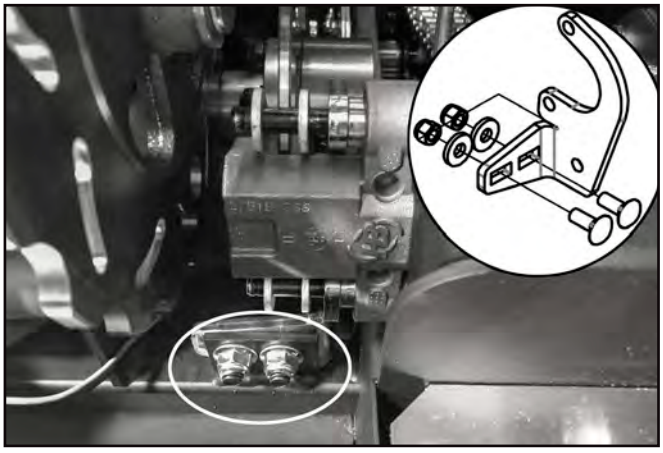
Replace the pads when:

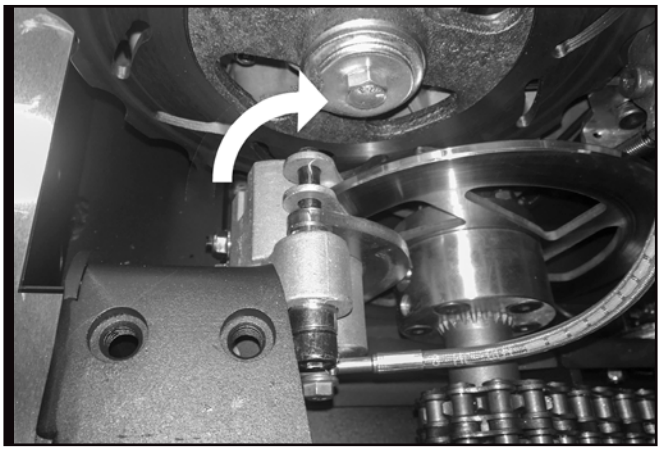
- the brake lining material moulded to each metal backing plate is worn to 0.035" (1/32”) thickness.
- the pads are glazed and brake performance is affected.
- the pads are contaminated with lubricant, and brake performance is affected.

Replacing The Service Brake Pads (Aurora Models Only)

1. Perform the Servicing Procedure, **Removing the Firewall**
2. Locate and remove the two (2) Nylon Locknuts that secure the Service Brake Caliper Bracket to the lower frame by (2) Carriage Bolts.

3. Rotate the caliper up and past the ROPS mounting bracket to access and remove the hex head bolts securing the pads in the caliper.
4. Slip out the brake pads from the caliper.





Installing The Service Brake Pads

CAUTION

Ensure the brake pads are NOT installed backwards. The pad side must be installed towards the brake disc. The brake pads utilize a black coloured anti-squeal coating on the opposite side of the pad which could be mistaken for the pad surface.

1. Before installing the new pads, gently pry the pistons back into the calliper first to allow clearance for the new pads. Slip the new pads into the calliper and reinstall bolts after applying Blue 242 LOCTITE to the threads. **Torque to 25 FT.LBS. 20.3Nm**
2. Attach the calliper and bracket back to the vehicle main frame using two (2) C/BOLT 3/8-16X1.0 GR5 ZN, WASHER .39X.94X.13 ZN Di-Chro and Locknut Nylon 3/8-16 ZP.

IMPORTANT

Loosely snug the nuts to the brake bracket. Apply the hand brake a few times to set the brake bracket position, then apply the brake and hold while tightly securing the nuts. Torque to 31 FT.LBS

Service Brake Bedding In Procedure

Bedding in of the service brakes is recommended by the brake pad manufacturer to obtain maximum performance and wear from the brake pads. When new service brake pads have been installed, the following procedure will need to be completed. Please include this procedure immediately after service pads have been replaced.

1. Ensure that the service brake system has been bled to provide maximum braking.
2. For this procedure select an area that is open and flat, such as an empty parking lot.

3. Drive the Argo at a speed of approximately 20kph. While continuing to apply throttle, gradually apply the service brake, slowing your speed down to 10kph, then release the brake and accelerate back to 20kph. Do not bring the Argo to a full stop (unless required for safety reasons). **Bringing the ARGO to a full stop when the brakes are hot may cause the brake pad to imprint itself on the rotor. If this happens it will cause vibration and poor brake performance.**
4. Repeat step 3 a total of 10 times. Do not wait between cycles to let brakes cool.

CAUTION

Brake components will be extremely hot at this point.

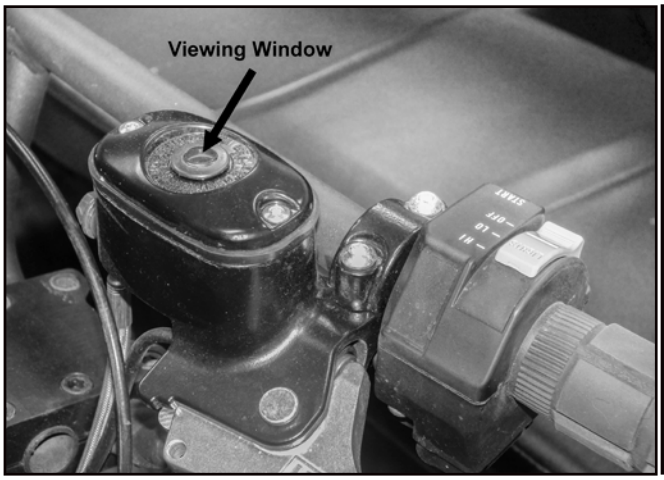
After the 10 cycles are complete, shut down the ARGO and allow the brakes to cool down. When brakes have cooled, repeat steps 3 and 4. The bedding procedure is now complete. The brakes may smell, and some smoke could be present.

Checking Hydraulic Brake Fluid Level (Handle Bar Master Cylinder)

All models are equipped with a hydraulic hand brake system. The master cylinder is mounted on the left side steering bar with a fluid level viewing window.
Monitor the hand brake fluid on a regular basis. If fluid level needs replenishing, remove the cover and ensure the fluid is to the “top” level mark.

IMPORTANT

If the cover needs to be removed to replenish or service the system, thoroughly clean the cover and surrounding area before removing to avoid any contamination to the brake system. Ensure the master cylinder is level before removing the cover. This may require loosening the master cylinder's clamp screws and rotating the master cylinder until it is level to the ground. Torque cover screws 6-8 in-lb (.67-.90 Nm)





Inspect all brake hoses and brake fittings at both hand brake and hydraulic calipers for any signs of brake fluid leaks.

Brake Fluid Inspection

WARNING

Do not allow brake fluid to contact your eyes, clothes, or painted surfaces. Brake fluid is extremely corrosive and can cause severe personal injury. Wear protective eye goggles, gloves and clothing. If brake fluid contacts your eyes, consult a physician immediately.

CAUTION

Regular inspection of the brake fluid is very important. Low fluid levels can cause a reduction or loss in braking ability; high fluid levels, or brake fluid contaminated with oil, dirt, or water, can damage the seals or other component parts of the hydraulic brake system.

Inspecting The Brake Fluid Condition (Steering Brake Master Cylinders)

Maintenance Schedule- Inspect every 50 hours of vehicle operation.

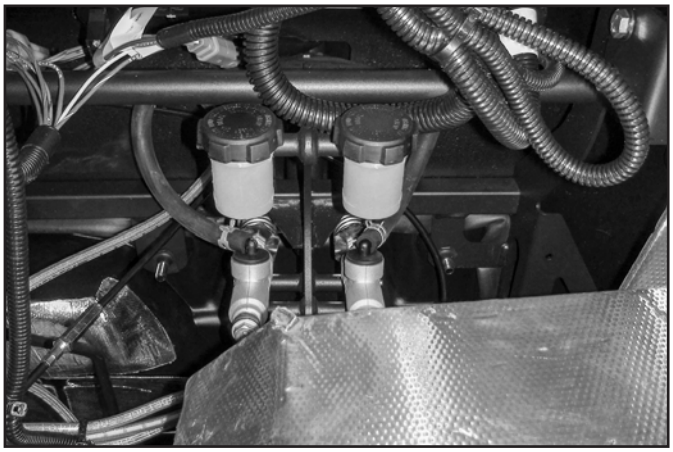
Use only Dot 4 brake fluid. Other brake fluids may not be compatible with ARGO brake components and operating temperatures.

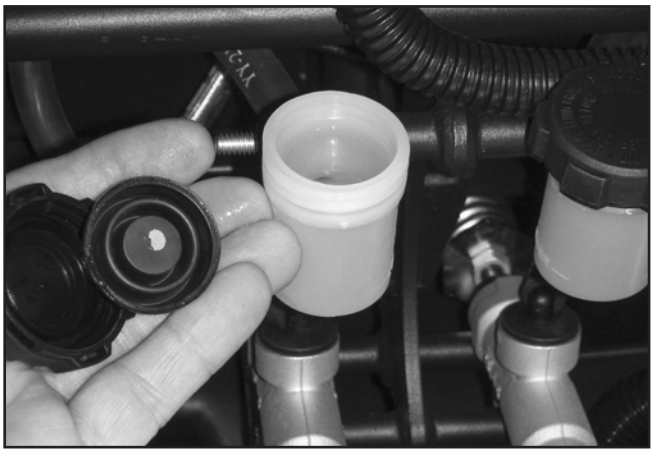
1. Locate the master cylinders in the engine compartment.

NOTE

To access the master cylinder cover, remove the hood assembly.

2. Clean the casing and lid of the master cylinder before removing. This will prevent dirt or other contaminants from entering the reservoir.
3. Thread the master cylinder covers off counter-clockwise and remove the cylinder lid and gasket.





Inspect the condition of the brake fluid and replace if:

- the fluid appears cloudy
- the fluid appears discoloured
- the fluid contains particles of dirt, rust, water, or other contaminants
- the level is below the viewing window

If any of these conditions are noted, perform the servicing procedure, **Changing the Brake Fluid** below.

Changing The Brake Fluid (Steering Brakes)

CAUTION

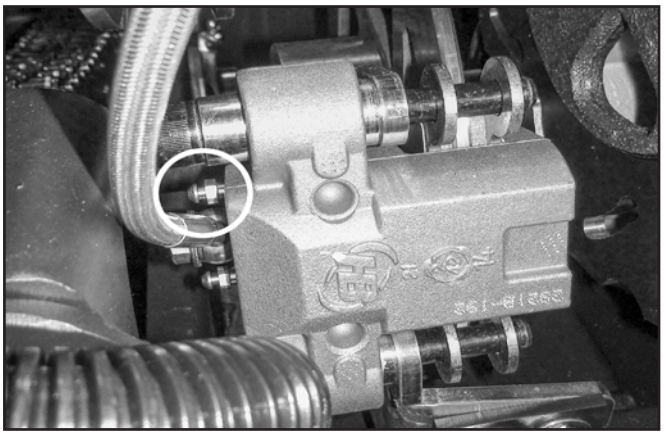
Clean the master cylinder casing and fittings. This will prevent dirt from contaminating the brake system. Contaminants in the brake fluid can damage components or cause premature wear to them. This will result in a loss of braking effectiveness. Ensure proper disposal of old brake fluid in accordance with local regulations.

1. To drain and replenish the brake system you will need to remove the firewall of the vehicle. Perform the servicing procedure, **Removing The Firewall** in this service guide.
2. Remove the engine hood to access the master cylinders and remove both covers.
3. Locate the bleed screw on the hydraulic brake caliper. Attach a transparent hose to the bleed screw and run to a container for disposal later. Slowly loosen the brake caliper bleed screw and gently pump the steering bar to purge the brake fluid from the system.
4. Continue to pump the fluid from the system while monitoring the master cylinder. Repeatedly replenish the master cylinder while pumping the old fluid from the system.
5. When the old fluid at the bleed screw appears to be replaced by fresh fluid, perform the servicing procedure, **Bleeding the Brakes (Steering Brakes)**.
6. Add brake fluid to the master cylinder level indicated on the reservoir.

Changing The Brake Fluid (Service Brakes) Aurora and 600 Models

1. Remove the master cylinder lid on the left side steering bar. Ensure the master cylinder is level before removing the cover. This may require loosening the master cylinder's clamp screws and rotating the master cylinder until it is level to the ground.

2. Locate the bleed screw on the hydraulic service brake caliper. Attach a transparent hose to the bleed screw and run to a container to collect the used fluid for disposal later. Slowly loosen the brake caliper bleed screw and gently squeeze the hand brake lever to purge the brake fluid from the system.



3. Continue to pump the fluid from the system while monitoring the master cylinder. Repeatedly replenish the master cylinder while pumping the old fluid from the system.
4. When the old fluid at the bleed screw appears to be replaced by fresh fluid, perform the servicing procedure, **Bleeding The Brakes (Service Brakes**)
5. Add brake fluid to the master cylinder level indicated on the reservoir.
6. Replace master cylinder lid and **torque cover screws 6-8 in-lb (.67-.90 Nm).**

Bleeding The Hydraulic Brake System (Service Brakes)

1. Remove master cylinder cover and add DOT 4 (ONLY) brake fluid as necessary.
2. With the bleed screw still tight, purge any air from the system by gently pumping the brake lever. When there are no longer visible signs of air bubbles observed in the master cylinder and some resistance is felt in the handbrake, replenish it with more fluid to bring it back up to the top of the full level line.



3. Next, apply steady pressure to the brake lever until the brake is fully engaged.
4. Hold the brake lever in this position and slowly loosen the caliper bleed screw 1/4 turn. Open the bleed screw only enough to slowly release fluid out of the screw. If there is any air still present in the system, it will be noticeable through this screw with a slight “spitting” of fluid. Re-tighten the screw and once again pump the brake system *slowly* a few more times. Apply pressure to the brake lever again and hold. Open the bleed screw gradually until fluid slowly releases. Repeat this procedure until the fluid coming from the bleed screw is a full stream with no visible signs of air in the system.

NOTE

Catch any purged fluid at the caliper bleed screw with a rag or small container. Clean up any residual brake fluid with a brake cleaner.

5. Replenish the master cylinder to the top level mark and secure the cover. **Torque cover screws 6-8 in-lb (.67-.90 Nm)**
6. Pump the brake lever a few times to build up proper pressure and locate the pads in the caliper assembly.

Hydraulic Brake Calipers
Maintenance Schedule - Inspect every 50 hrs. of vehicle operation

CAUTION

The brakes must be inspected regularly by a qualified authorized Argo dealer/Technician. Damaged or leaking calipers can seriously impair the ARGO steering/braking capability. Severe personal injury or property damage could result from a damaged or poorly adjusted brake system.

Inspect the calipers for the conditions listed in the following chart and then perform the recommended servicing procedures.

Condition	Corrective Action	Servicing Procedure	Page
Cracked or Broken Calipers	Replace Immediately	Hydraulic Brake Caliper Removal	BR-9
Broken or Missing	Replace Immediately bolts	Hydraulic Caliper Installation	BR-10
Hydraulic Fluid Leakage	Inspect Caliper Casting and “O” Ring	Hydraulic Brake Caliper Removal	BR-9
	Inspect Hoses and Connections	Inspecting the Hydraulic Brake line Hydraulic Caliper Installation	BR-17 BR-10

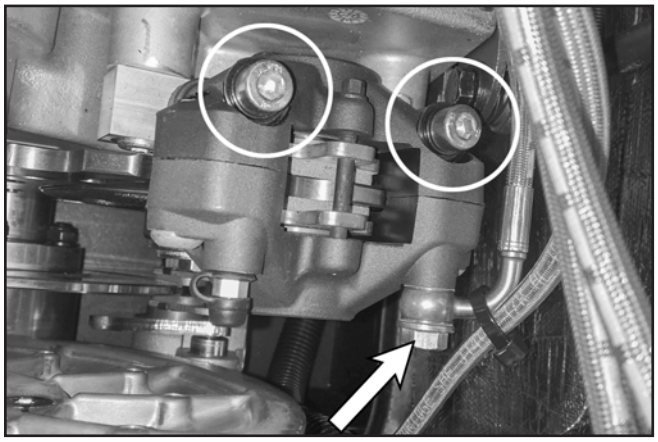
Removing The Hydraulic Brake Caliper - Aurora Steering Brakes

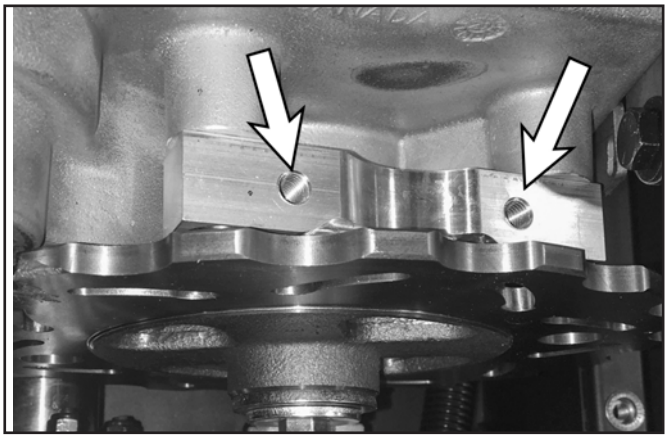
1. Perform the servicing procedure, **Removing The Firewall**

WARNING

Use the appropriate PPE when working with hydraulic brake fluid. Do not allow brake fluid to contact your eyes, skin or clothes. If brake fluid contacts your eyes, consult a physician immediately.

2. Disconnect the brake line at the caliper. Be aware of the brake fluid present in the system. Use a rag or small container to collect any initial fluid that may drain from the master cylinder until you can raise and secure the brake line high enough to stop the flow of fluid.
3. Remove the two (2) SHCS,M8x1.25x65 Gr12.9 Znm, two (2) Lockwasher 5/16 Split-Ring Bnz and two (2) Washer .34x.70x.06 Zn bolts securing the steering calipers to the transmission housing.





Reinstalling The Hydraulic Steering Brake Caliper - Aurora Models

1. Install the brake line to the steering brake caliper.
2. Secure the steering brake caliper to the transmission with two (2) SHCS,M8x1.25x65 Gr12.9 Znm, two (2) Lockwasher 5/16 Split-Ring Bnz and two (2) Washer .34x.70x.06 Zn. **Torque to 283 in.lbs. (31.9Nm.)**
3. **Torque the brake line banjo bolt to 20-22 FT LBS. (27.12 Nm)**.
4. Perform the servicing procedure, **Bleeding The Hydraulic Brake System (Service Brakes)**

Bleeding the Brakes (Steering Brakes)

1. Remove master cylinder covers and add DOT 4 (ONLY) brake fluid as necessary.
2. Remove the rubber plug from the bleed screw at the Steering Brake Caliper.

3. With the bleed screw still tight, purge any air from the system by pumping the steering handles slowly. When there are no longer visible signs of air bubbles observed in the master cylinder, replenish it with more fluid to bring it back up to the top level mark.
4. Next, apply steady pressure to the steering lever until the brake is fully engaged. Hold the steering lever in this position and slowly loosen the caliper bleeding screw 1/4 turn. Open the bleed screw only enough to slowly release fluid out of the screw. If there is any air still present in the system, it will be noticeable through this screw with a slight "spitting" of fluid. Re-tighten the screw and once again pump the brake system *slowly* a few more times. Apply pressure to the steering lever again and hold. Open the bleed screw gradually until fluid slowly releases. Repeat this procedure until the fluid coming from the bleed screw is a full stream with no visible signs of air in the system.

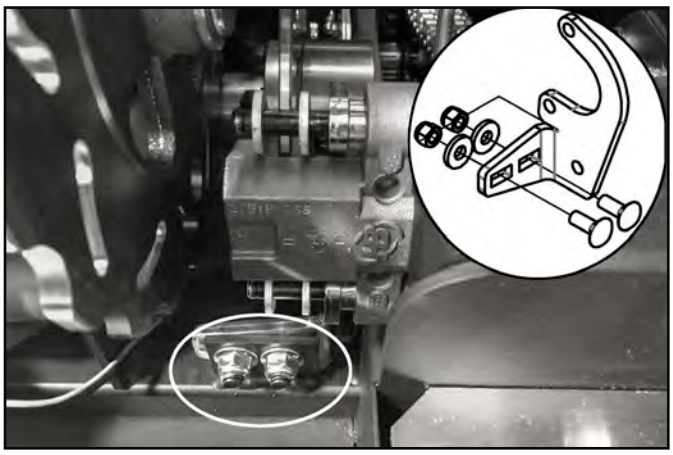
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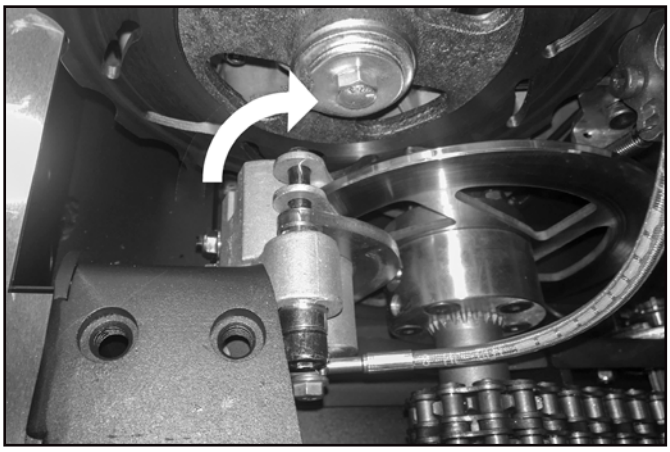
Catch any purged fluid at the caliper bleed screw with a rag or small container.

5. Replenish the master cylinder to the full level mark and secure the cover and rubber gasket.
6. Clean any stray brake fluid that might have found its way on to the brake disc with brake cleaner. Brake fluid on the brake disc can seriously affect braking capabilities of the ARGO vehicle.
7. Pump the steering levers a few times to build up proper pressure and locate the pads in the caliper assembly.

Remove Hydraulic Brake Caliper - Service Brakes (Aurora)

1. Perform the Servicing Procedure, **Removing The Firewall**
2. Disconnect the brake line at the caliper. Be aware of the brake fluid present in the system. Use a rag or small container to collect any initial fluid that may drain from the master cylinder until you can raise and secure the brake line high enough to stop the flow of fluid.
3. Locate and remove the two (2) 9/16" Nylon Locknuts that secure the Service Brake Caliper Bracket to the lower frame by (2) Carriage Bolts. *See following Page.*
4. Rotate the caliper up and past the ROPS mounting bracket. *See following page.*
5. Remove the caliper and pad assembly from the caliper mounting bracket.





Installing The Hydraulic Brake Caliper - Service Brakes (Aurora)

1. Attach the brake line to the new caliper and **torque to 20-22 FT LBS. (27.12 Nm)**,orienting the hose in the same direction as previously installed.
2. Re-attach the replacement caliper to the mounting bracket and slip the pads over the brake disc.
3. Rotate the caliper and bracket assembly back down to the vehicle main frame and re-attach it using two (2) C/Bolt 3/8-16X1.0 GR5 ZN, Washer .39X.94X.13 ZN Di-Chro and Locknut, Nylon 3/8-16 ZP.

IMPORTANT

Loosely snug the nuts. Apply the hand brake a few times to set the brake bracket position, then apply the brake and hold while tightly securing the nuts. Torque to 31 FT.LBS

Removing the Hydraulic Brake Caliper - Frontier Steering Brakes

IMPORTANT

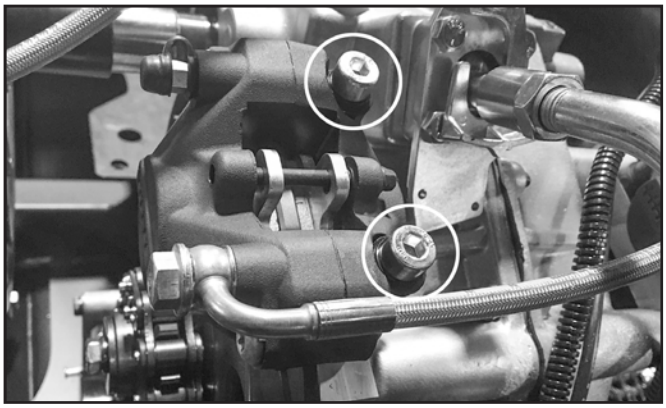
1. Perform the servicing procedure, **Removing The Firewall**

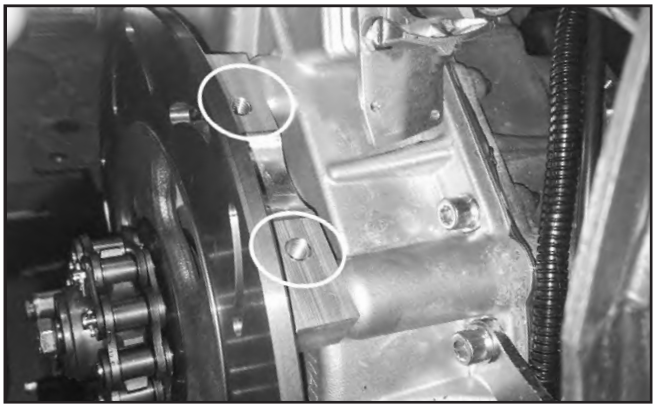
WARNING

Use the appropriate PPE when working with hydraulic brake fluid. Do not allow brake fluid to contact your eyes, skin or clothes. If brake fluid contacts your eyes, consult a physician immediately.

2. Disconnect the brake line at the caliper. Be aware of the brake fluid present in the system. Use a rag or small container to collect any initial fluid that may drain from the master cylinder until you can raise and secure the brake line high enough to stop the flow of fluid.

3. Remove the two (2) SHCS,M8x1.25x65 Gr12.9 Znm, two (2) Lockwasher 5/16 Split-Ring Bnz and two (2) Washer .34x.70x.06 Zn bolts securing the steering calipers to the transmission housing.
4. Remove the Steering Brake Caliper using a 6mm hex head socket wrench.





Reinstalling The Hydraulic Steering Brake Caliper - Frontier Models

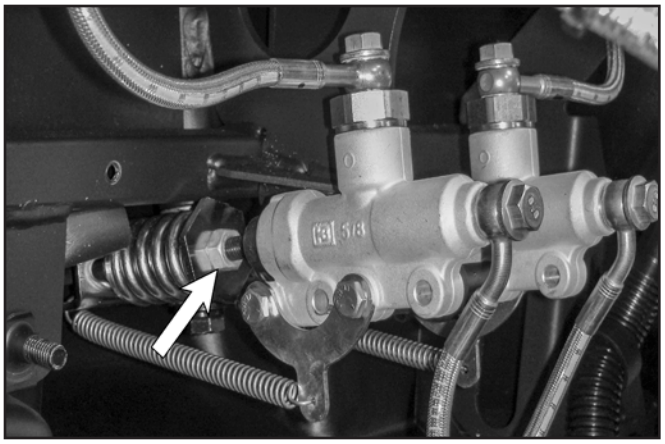
1. Install the brake line to the steering brake caliper.
2. Secure the steering brake caliper to the transmission with two (2) SHCS,M8x1.25x65 Gr12.9 Znm, two (2) Lockwasher 5/16 Split-Ring Bnz and two (2) Washer .34x.70x.06 Zn. **Torque to 283 in.lbs. (31.9Nm.)**
3. **Torque the brake line banjo bolt to 20-22 FT LBS. (27.12 Nm)**.
4. Perform the servicing procedure, **Bleeding the Brakes (Steering Brakes)**

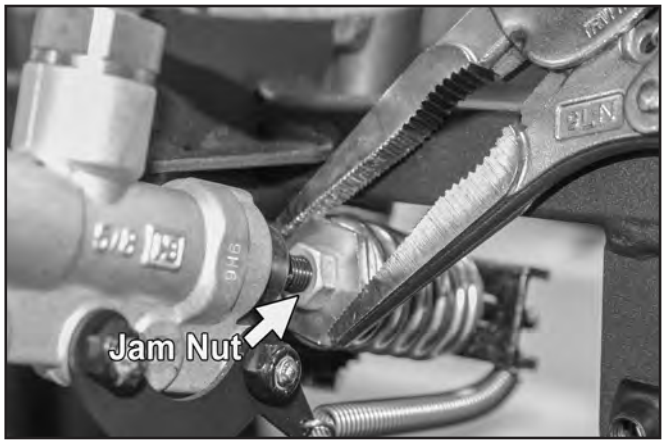
Hydraulic Brake Plunger Pin Adjustment

IMPORTANT

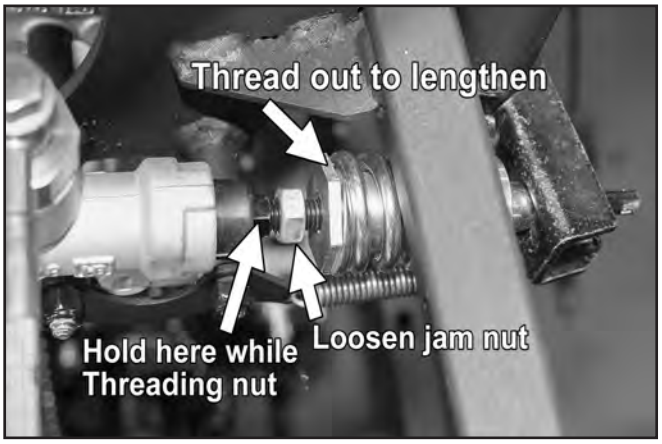
It is critical that the master cylinder pistons are adjusted properly when the steering handlebars are in the centred position. Overheating of the brake system could occur due to the piston being adjusted too far in. This could cause a drag on the system and a possible brake lockup or brake fade. The piston being adjusted too far out increases the distance the piston is required to travel in order to provide brake pressure.

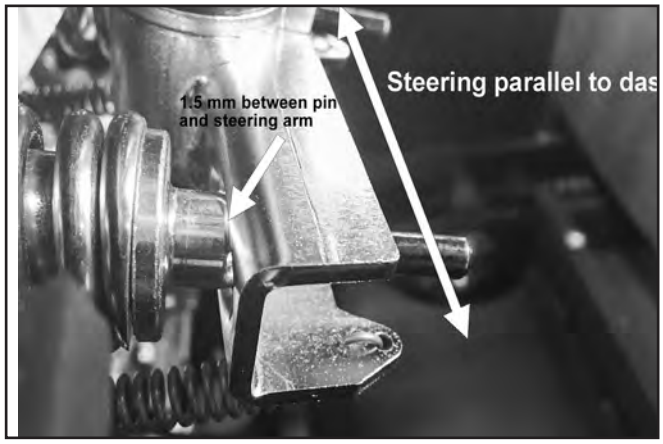
1. Remove the hood of the vehicle to locate the steering plunger pin assembly.
2. Loosen the jam nut with a 10mm wrench while securing the larger nut with another wrench or needle nose visegrip. *See following page.*





3. While holding the small nut on the plunger piston, thread the larger nut out to lengthen the plunger pin, or in to shorten it.
4. Ensuring that the steering bar is parallel with the dash of the vehicle, adjust the plunger pin until it comes close to contacting the steering arm. Leave a slight gap of about 1.5mm between pin contact of steering arm.





5. Re-tighten the jam nut.

Hydraulic Brake Hose Maintenance Schedule

Inspect every 25 hrs. of vehicle operation.

1. Inspect the condition of the hydraulic hoses. Replace them if:
   - the hoses are cracked or punctured
   - brake fluid leakage is noted on hose, nearby assemblies, or around hose fittings
2. Perform the servicing procedure, **Replacing the Hydraulic Brake Hoses,** below

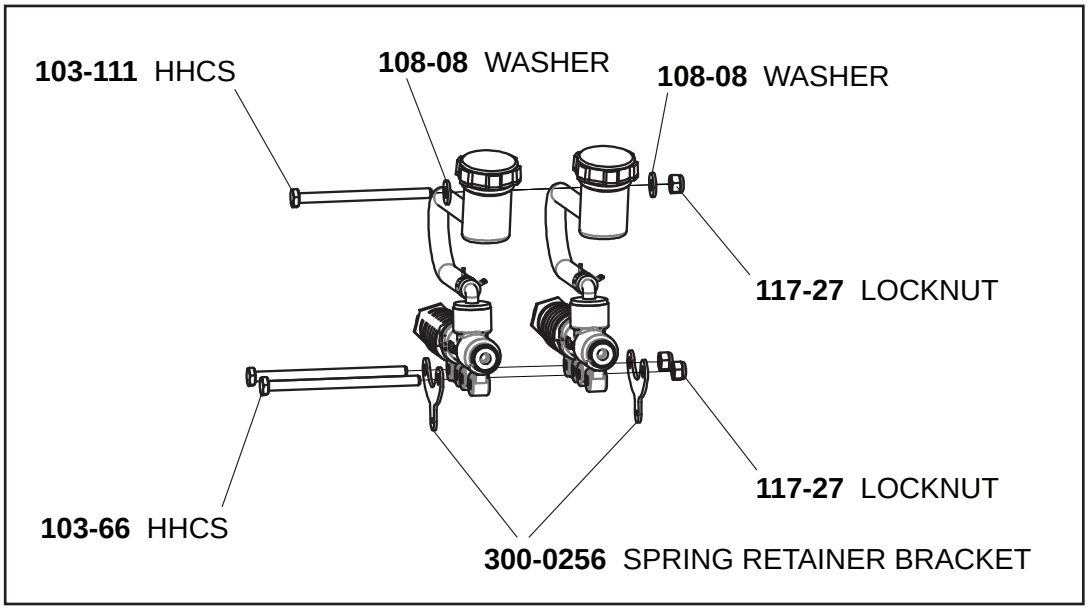
Replacing the Hydraulic Brake Hoses

IMPORTANT

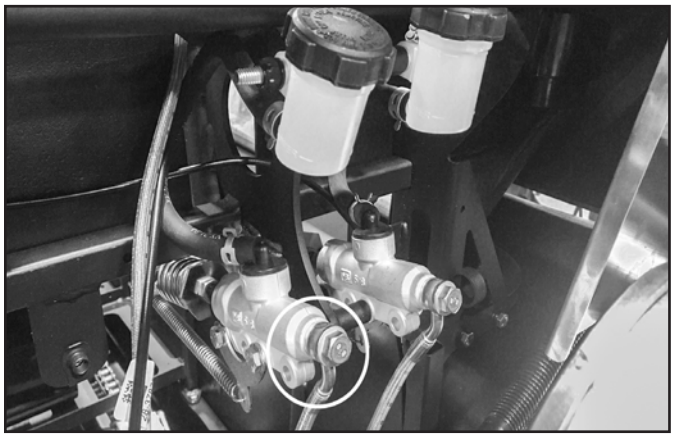
Clean the master cylinder casing and the caliper casting and fittings. This will prevent dirt from contaminating the brake system. Contaminants in the brake fluid can damage components or cause a loss in braking or steering effectiveness.

1. Remove cover and gasket seal from the master cylinder reservoir.
2. Disconnect the hose from the caliper. Place a catch tray under the free end of the hydraulic hose and allow the hose and master cylinder reservoir to drain.
3. Disconnect the hydraulic hose from the RMCI assembly.
4. Replace the hose.
5. Connect the new hydraulic hose to the master cylinder and RMCI assembly. Make sure both copper washer seals have been replaced with new ones at the banjo bolts. Ensure the fittings and hoses are oriented and tie-wrapped in the same manner as they were previously installed.
6. **Torque the brake line banjo bolt to 20-22 FT LBS. (27.12 Nm)**.
7. Perform the servicing procedure, **Bleeding the Hydraulic Brake System (Service Brakes)**

Removing The RMCI Assembly (Aurora & EU 600 Models)



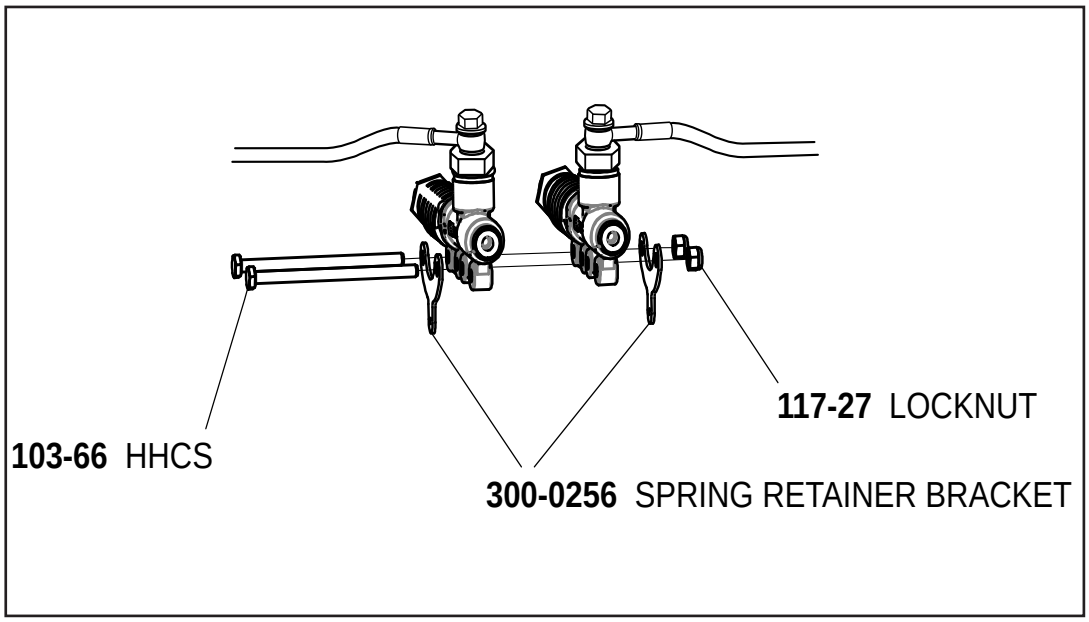
1. Remove the vehicle hood and locate the Master Cylinder/Plunger Pin assembly.
2. Disconnect the hose from the caliper. Place a catch tray under the free end of the hydraulic hose and allow the hose and master cylinder reservoir to drain.
3. Remove the HHCS 1/4-20x4.0 GR5 Zn & Locknut, Nylon 1/4-20 Zn securing the RMCI Assembly to the steering column frame. Be careful of washers and the Spring Bracket Retainers used in the assembly.



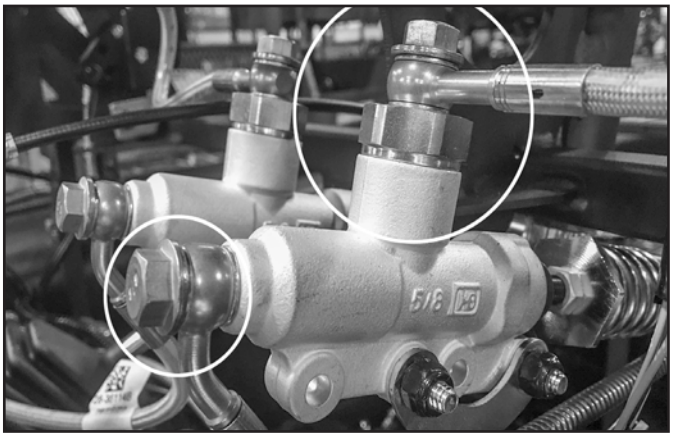
Installing The RMCI Assembly(Aurora & EU 600 Models)

1. Install the replacement RMCI Assembly and **torque nuts to 76 in.lbs.(8.58 Nm).**
2. Reconnect the brake line installing two (2) new copper washers. **Torque to 20-22 FT LBS. (27.12 Nm)**.
3. Fill the master cylinder with fresh DOT 4 brake fluid.
4. Perform the servicing procedure, **Bleeding The Hydraulic Brake System (Steering Brakes).**
5. Re-check the brake plunger pin adjustment and perform the servicing procedure, **Hydraulic Brake Plunger Pin Adjustment** if necessary.

Replacing The RMCI Assembly (Frontier Models)



1. Remove the vehicle hood and locate the Master Cylinder/Plunger Pin assembly.
2. Disconnect both brake lines at the caliper. Be aware of the brake fluid present in the system. Use a rag or small container to collect any initial fluid that may drain from the RMCI or the master cylinder which is located on the left hand steering bar of the vehicle.



3. Remove the HHCS 1/4-20x4.0 GR5 Zn & Locknut, Nylon 1/4-20 Zn securing the RMCI Assembly to the steering column frame. Be careful of washers and Bracket, Spring Retainers used in the assembly.

Installing The RMCI Assembly (Frontier Models)

1. Install the replacement RMCI Assembly and **torque nuts to 76 in.lbs.(8.58 Nm)**
2. Reconnect the brake lines installing two (2) new copper washers at each fitting. **Torque to 20-22 FT LBS. (27.12 Nm)**.
3. Fill the master cylinder with fresh DOT 4 brake fluid.
4. Perform the servicing procedure, **Bleeding The Hydraulic Brake System (Steering Brakes).**
5. Re-check the brake plunger pin adjustment and perform the servicing procedure, **Hydraulic Brake Plunger Pin Adjustment** if necessary.

Brake Disc Inspection

Maintenance Schedule - Inspect every 50 hrs. of vehicle operation.

IMPORTANT

Badly worn brake pads, or abnormal use of the vehicle, can cause serious damage to the brake discs. Inspect the brake discs for the condition listed in the chart below, and perform the recommended servicing procedures.

Always refer to the PARTS MANUAL that corresponds to the specific model and serial number indicated on the serial tag plate of your vehicle. This is essential to locating and ordering the correct components that were used during that particular manufacturing period.

Brake Disc Condition	Corrective Action	Servicing Procedure
Rust or pitting Bends or cracks Scratches or scoring Broken or worn sprocket teeth	Replace Immediately	1. Brake Disc Removal 2. Brake Disc Installation
Glazing Oil or grease contamination	Clean	1. Brake Disc Removal 2. Brake Disc Installation

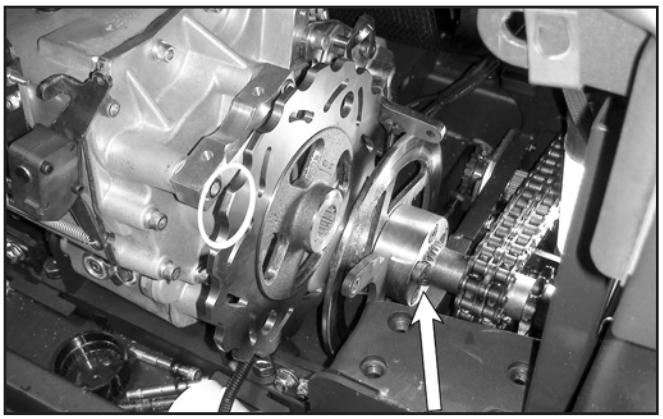
Removing The Steering Brake Disc (Aurora Models)

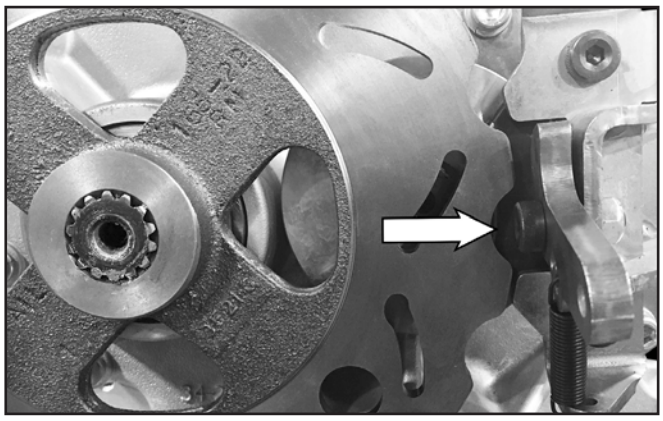
1. Perform the servicing procedure, **Removing The Firewall.**
2. Perform the servicing procedure, **Remove Hydraulic Brake Caliper - Steering Brakes**
3. Perform the servicing procedure, **Remove Hydraulic Brake Caliper - Service Brakes (Aurora)**

NOTE

If removing the LEFT side brake disc you will need to remove both Drive Belt and Driven Clutch to successfully work on and remove the brake disc. Brake lines do NOT have to be disconnected from the brake calipers.

4. Remove the HHCS 3/8-24 x 1 Gr5 and all washers securing the brake disc to the transmission output shaft. When removing the brake discs be aware of any shims used in behind the brake disc and ensure they are reinstalled with the new brake disc.
5. Using a 1/2" open end wrench and 3/16" hex head allen wrench, remove one (1) nut from the Service Brake Disc coupler bolt using and back it out from the coupler.
6. Rotate steering brake disc until one of the outer convex edges of the brake disc aligns with the backed out coupler bolt.
7. Ensure one of the inner concaves of the Steering Disc Brake is aligned with the parking brake lever bolt. Slide the Steering brake disc from the shaft and out of the vehicle.





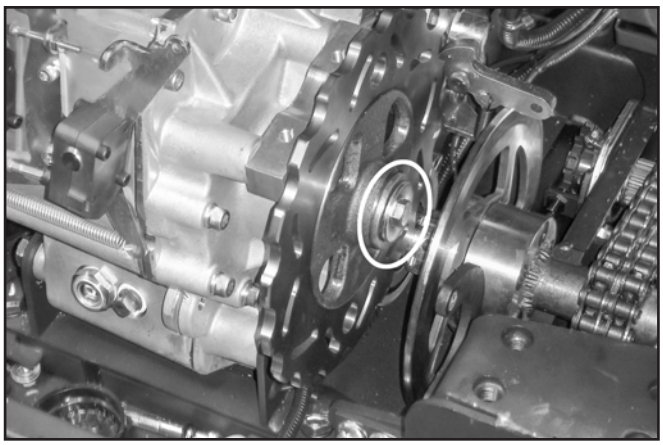
Installing The Steering Brake Disc (Aurora Models)

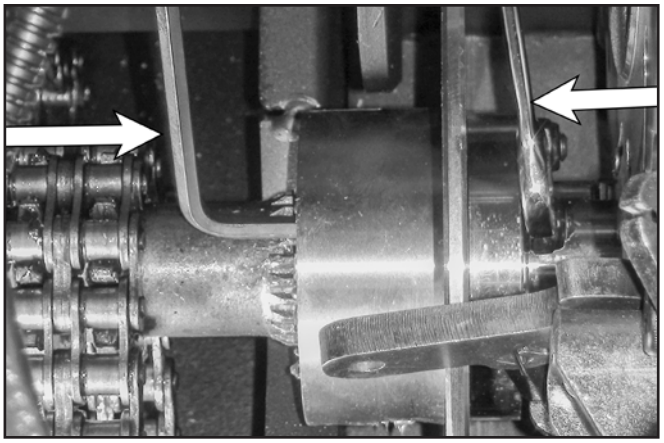
1. Perform the servicing procedure, **Brake Disc Inspection.**
2. Apply some anti-seize compound to the Steering Brake Disc Shaft splines before reinstalling the brake disc.
3. Install the brake disc to the shaft ensuring any spacers previously installed *behind* it are installed first. Ensure brake disc profile is aligned properly with idler shaft coupler mounting bolt and parking brake mounting bolt as described in the removal step above.

IMPORTANT

The brake disc should slide on by hand. Do NOT force or hammer to install.

4. Apply Blue 242 LOCTITE to the threads of the HHCS 3/8-24 x 1 Gr5 Bolt and secure the disc with all lockwashers and flat washers as previously installed.
5. **Torque the brake disc bolt to 30 FT.LBS. (40.6 Nm).** *See following page.*
6. Reinstall the coupler bolt nut remove in step 6 and **torque to 22 FT.LBS. (30.0 Nm)**





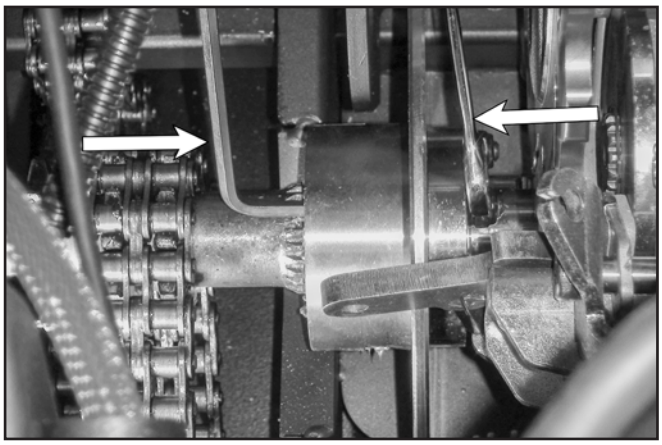
1. Perform the servicing procedure, **Installing The Hydraulic Brake Caliper - Service Brakes (Aurora)**
2. Perform the servicing procedure, **Installing The Hydraulic Brake Caliper - Steering Brakes**
3. Perform the servicing procedure, **Installing The Driven Clutch** (if the left side brake disc was serviced)
4. Perform the servicing procedure, **Installing The Drive Belt** (if the left side brake disc was serviced).
5. Perform the servicing procedure, **Installing The Firewall.**

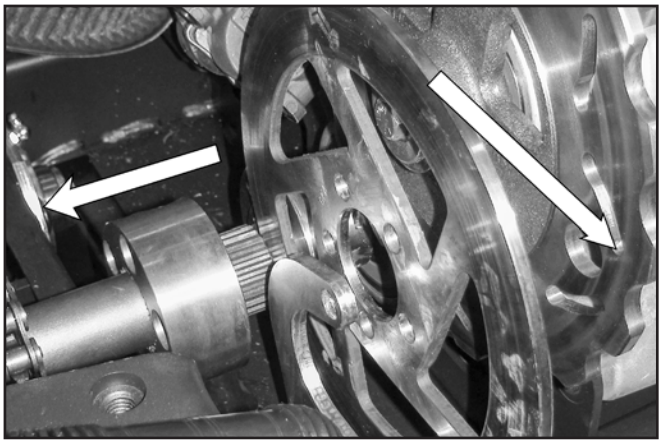
Removing The Service Brake Disc (Aurora Models Only)

NOTE

If you are removing the LEFT side Service Brake Disc you will need to remove both Drive Belt and Driven Clutch to successfully work on and remove the Service Brake disc. Brake line does NOT need to be disconnected from the caliper.

1. Perform the servicing procedure, **Removing The Firewall.**
2. Perform the servicing procedure, **Removing The Drive Belt** (if the left side Service Brake Disc is being serviced).
3. Perform the servicing procedure, **Removing The Driven Clutch** (if the left side Service Brake Disc is being serviced).
4. Perform the servicing procedure, **Removing The Hydraulic Brake Caliper - Service Brakes (Aurora)**
5. Remove all six (6) Bolt, Shoulder M10x45 securing the Service Brake at the coupler. They are secured by six (6) Lockwasher 5/16 Split-Ring Bnz and six (6) Locknut, Nylon M8x1,25mm Zn8. *See following page.*
6. Slide the coupler back on the output shaft and slip out the Service Brake Disc. *See following page.*





Installing The Service Brake Disc (Aurora Models)

1. Slip the replacement Service Brake Disc back to the output shaft and sandwich between the two couplers.
2. Align the holes in the Service Brake Disc with that of the couplers and reinstall the Bolt, Shoulder M10x45 securing with the Lockwasher 5/16 Split-Ring Bnz and Locknut, Nylon M8x1,25mm Zn8.
3. **Torque the coupler bolts to 22 FT.LBS. (30.0 Nm).**
4. Perform the servicing procedure, **Installing The Hydraulic Brake Caliper - Service Brakes (Aurora)**
5. Perform the servicing procedure, **Installing The Driven Clutch** (if the left side Service Brake Disc was serviced)
6. Perform the servicing procedure, **Installing The Drive Belt** (if the left side Service Brake Disc was serviced).
7. Perform the servicing procedure, **Installing The Firewall.**

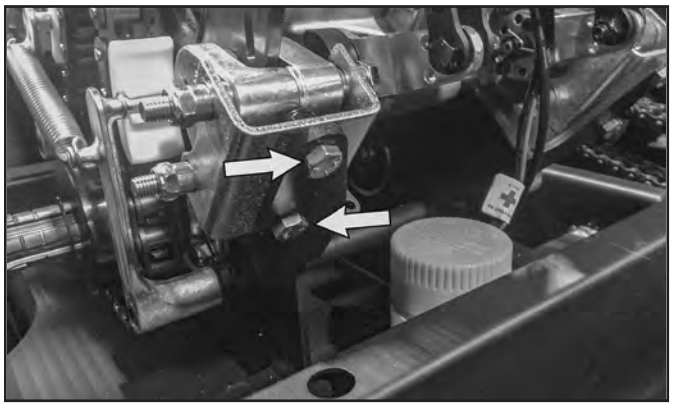
Removing The Steering Brake Disc (Frontier Models - Auto Idler Chain Adjust).

1. Perform the servicing procedure, **Removing The Front Firewall.**
2. Perform the servicing procedure, **Removing the Hydraulic Brake Caliper - Frontier Steering Brakes**

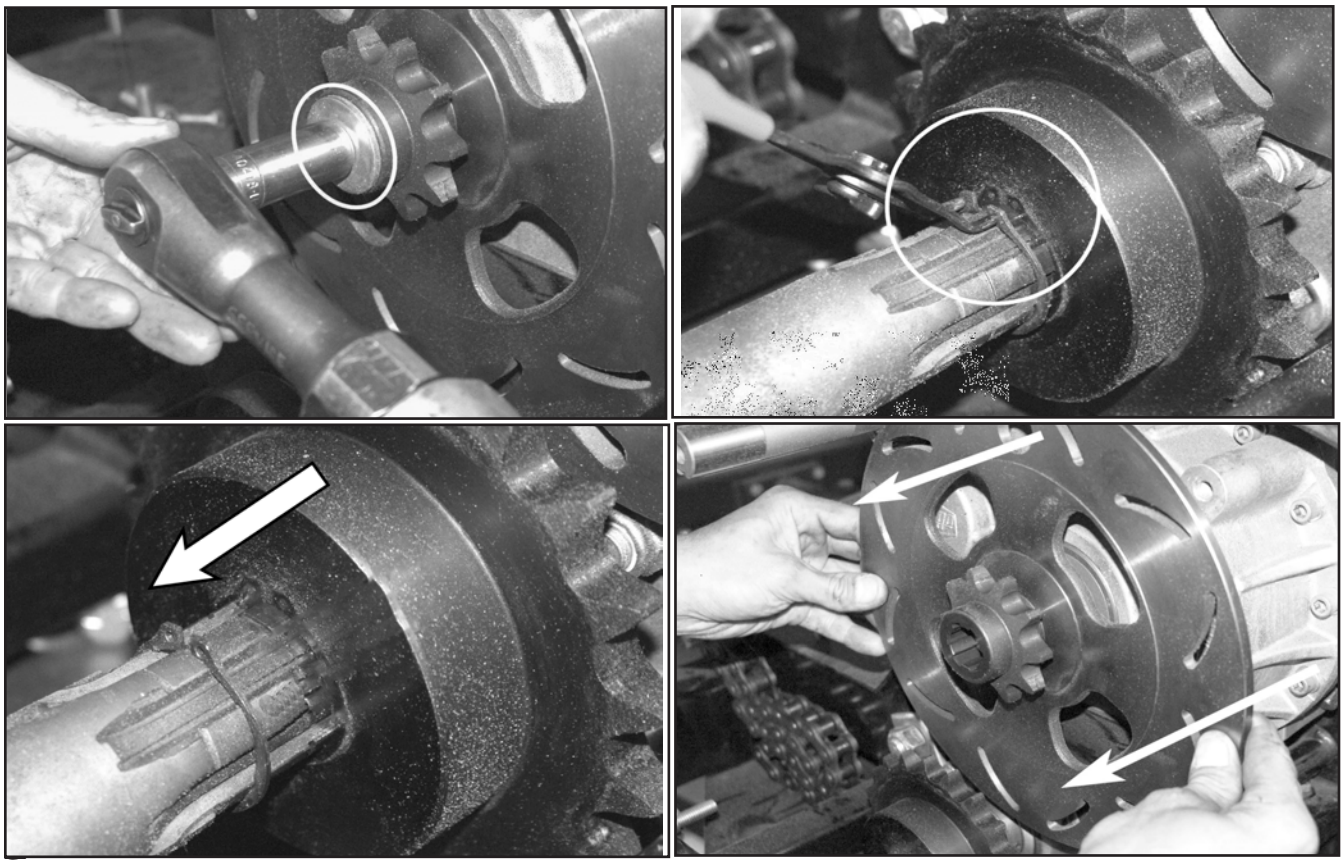
NOTE

If you are removing the LEFT side Service Brake Disc you will need to remove both Drive Belt and Driven Clutch to successfully work on and remove the Service Brake disc. Brake line does NOT need to be disconnected from the caliper.

3. Remove the Parking Brake Caliper from the caliper bracket using a 1/2" socket and a 1/2" open end wrench.



4. Perform the servicing procedure, **Removing The Idler Chain (Frontier Models - Auto Idler Chain Adjust).**
5. Using a 9/16" socket, remove the HHCS 3/8-24x1.25 Gr5 Zn securing the Steering Brake Disc with lockwasher and flat washer.
6. Pull back the retaining ring on the idler shaft drive sprocket. Slide the sprocket back against the retaining ring to allow clearance for slipping the Steering Brake Disc off the output shaft.
7. Remove the Steering Brake Disc.



8. Perform the servicing procedure, **Brake Disc Inspection**.

Installing The Steering Brake Disc (Frontier Models - Auto Idler Chain Adjust).

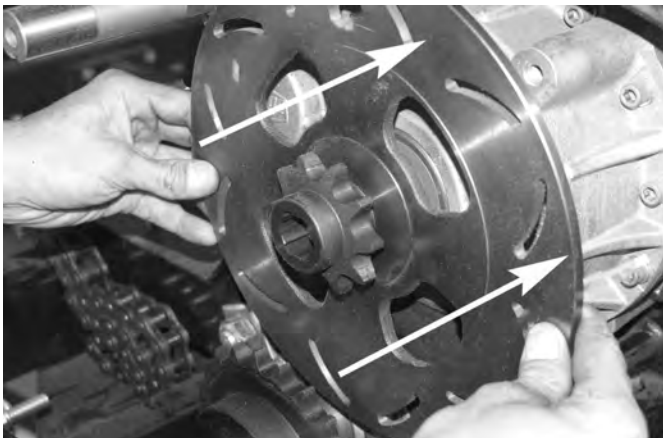
1. Before installing the Steering Brake Disc, inspect the splines of the output shaft for any damage.

2. Apply anti-seize compound to the output shaft spline.
3. Align the splines of the brake disc with those of the output shaft and slide the brake disc on to seat up against the shoulder of the shaft.

IMPORTANT

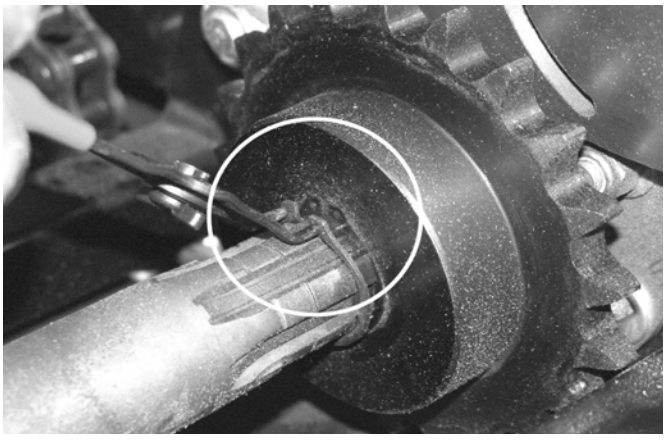
The brake disc should slide on by hand. Do NOT force or hammer to install.

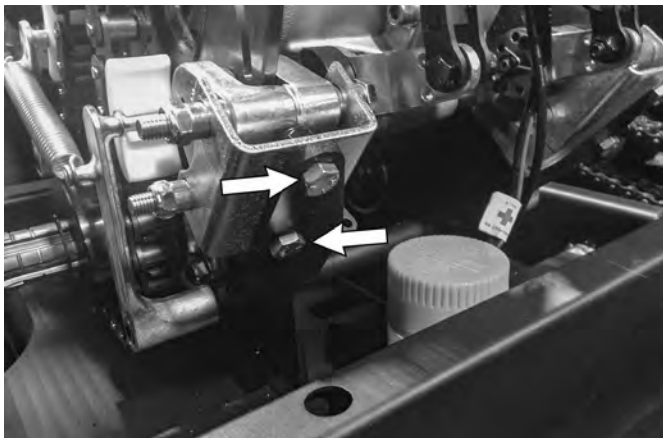
4. Using a 9/16" socket, tighten the HHCS 3/8-24x1.25 Gr5 Zn securing the Steering Brake Disc with lockwasher and flat washer.
5. Secure with the bolt and flat washer. **Torque to 30 FT.LBS (40.7Nm)**





6. Slide the idler shaft sprocket back into position and seat the retaining ring back to the original groove.
7. Perform the servicing procedure, **Installing the Idler Chain (Frontier Models - Auto Idler Chain Adjust).**
8. Reinstall the Steering Brake Caliper and **torque to 283 in.lbs. (31.9Nm).**
9. Reinstall the Parking Brake Caliper





10. Loosely snug the Parking Brake Caliper hardware. Squeeze the hand brake a few times to centre the pads and then hold the brake on while tightening the hardware to a final **torque of 209 in.lbs (23.6 Nm).**

11. Reinstall the firewall and secure with five (5) HHCS 5/16-18x1.25 GR5 Zinc and five (5) Washer .34x.70x.06 ZINC.

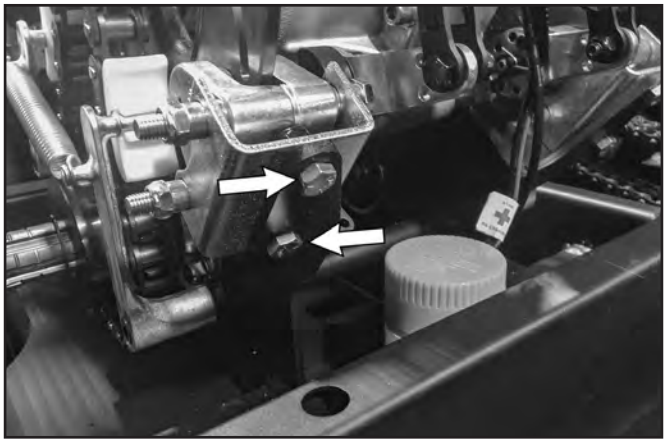
Removing The Steering Brake Disc (Frontier Models - Manual Idler Chain Adjust).

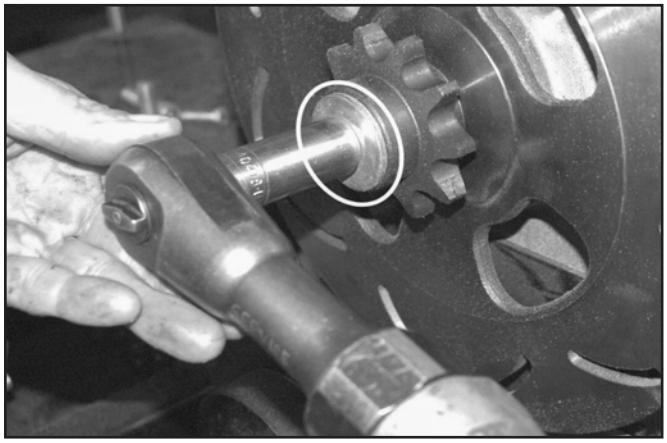
1. Perform the servicing procedure, **Removing The Front Firewall.**
2. Perform the servicing procedure, **Removing the Hydraulic Brake Caliper - Frontier Steering Brakes**

NOTE

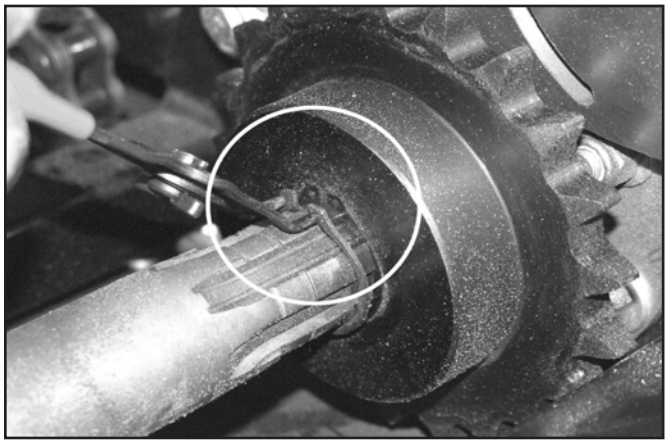
If you are removing the LEFT side Service Brake Disc you will need to remove both Drive Belt and Driven Clutch to successfully work on and remove the Service Brake disc. Brake line does NOT need to be disconnected from the caliper.

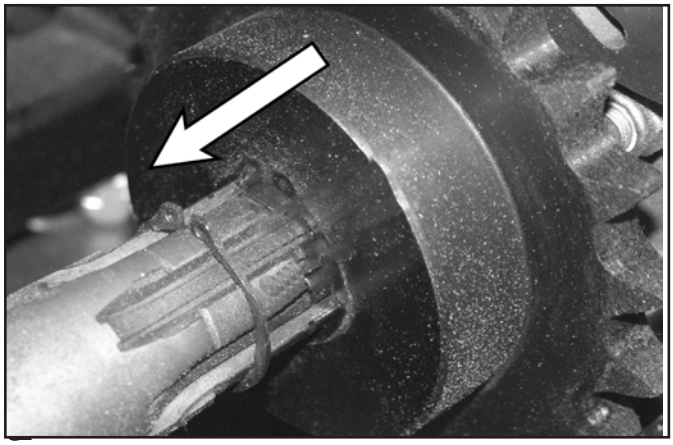
3. Remove the Parking Brake Caliper from the caliper bracket using a 1/2" socket and a 1/2" open end wrench.
4. Perform the servicing procedure, **Removing The Idler Chain (Frontier Models - Manual Idler Chain Adjust).**
5. Using a 9/16" socket, remove the HHCS 3/8-24x1.25 Gr5 Zn securing the Steering Brake Disc with lockwasher and flat washer.

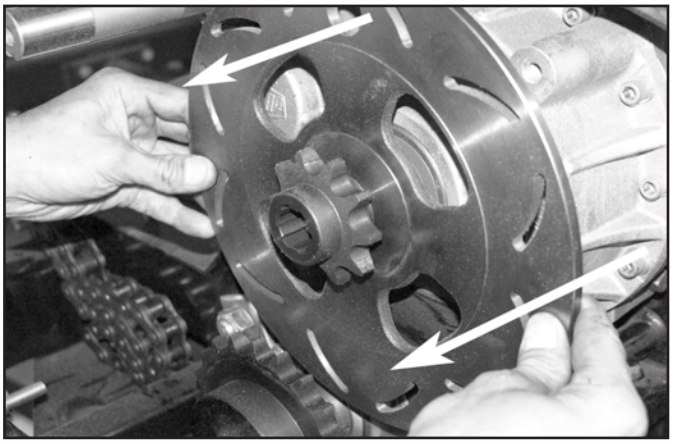




6. Pull back the retaining ring on the idler shaft drive sprocket. Slide the sprocket back against the retaining ring to allow clearance for slipping the Steering Brake Disc off the output shaft.







7. Remove the Steering Brake Disc.
8. Perform the servicing procedure, **Brake Disc Inspection**.

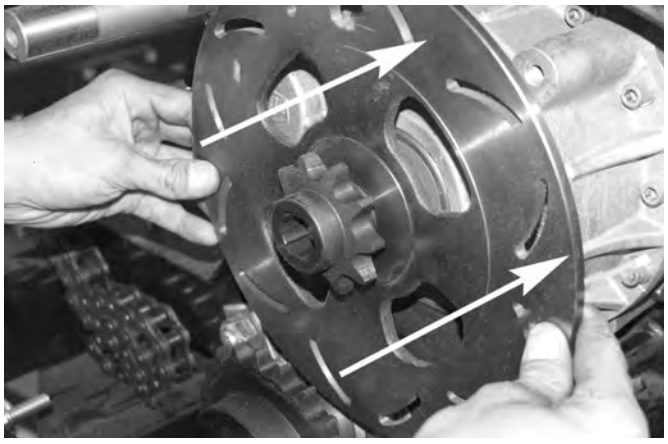
Installing The Steering Brake Disc (Frontier Models - Manual Idler Chain Adjust).

1. Before installing the Steering Brake Disc, inspect the splines of the output shaft for any damage.
2. Apply anti-seize compound to the output shaft spline.
3. Align the splines of the brake disc with those of the output shaft and slide the brake disc on to seat up against the shoulder of the shaft.

IMPORTANT

The brake disc should slide on by hand. Do NOT force or hammer to install.

4. Using a 9/16" socket, tighten the HHCS 3/8-24x1.25 Gr5 Zn securing the Steering Brake Disc with lockwasher and flat washer.
5. **Torque to 30 FT.LBS (40.7Nm)**



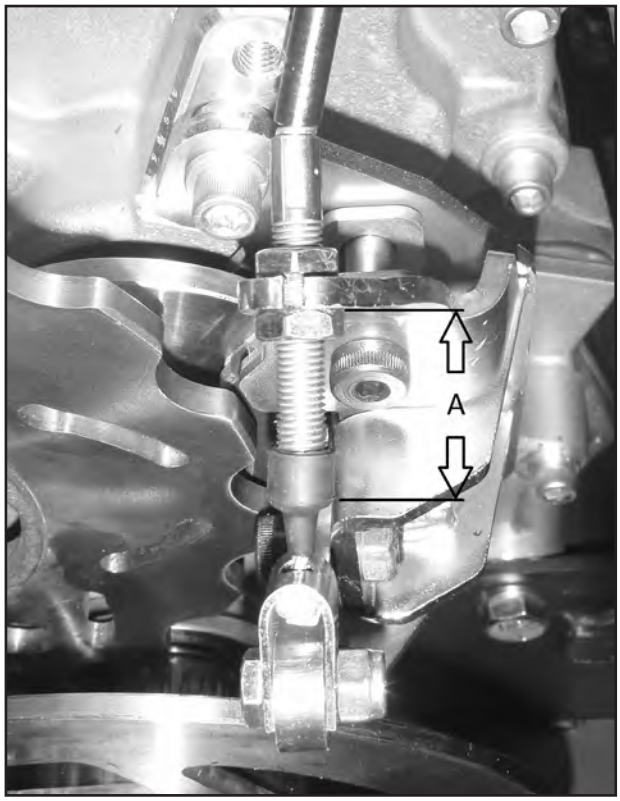


6. Slide the idler shaft sprocket back into position and seat the retaining ring back to the original groove.

7. Perform the servicing procedure, **Installing The Idler Chain (Frontier Models - Manual Idler Chain Adjust).**
8. Perform the servicing procedure, **Installing The Steering Brake Caliper.**
9. Perform servicing procedure, **Installing The Parking Brake Caliper.**
10. Perform the servicing procedure, **Installing the Firewall.**

Parking Brake System (Aurora Models)

Adjusting the Emergency/Parking Brake



There are 8 positions on the hand brake lever. The cable should be adjusted to have the 5th position (click) as fully engaged with normal firm effort (extra effort required for 6th). A properly adjusted cable with the hand lever fully down should allow the parking brake disk to turn easily between the pads while rocking the vehicle slightly back and forth. If the cable is too tight the brake will be engaged and the operator may experience a sluggish vehicle and cause the pads to wear prematurely. If the 5th position (click) does not provide full engagement with normal firm effort (extra effort for the 6th), adjust the cable accordingly. Loosen the jam nut at the caliper and adjust as needed. Reduce distance "A" to provide more braking force or increase distance "A" if brake does not turn freely.

WARNING

The use of an improperly adjusted emergency/parking brake is a serious hazard, and could lead to vehicle damage or personal injury.

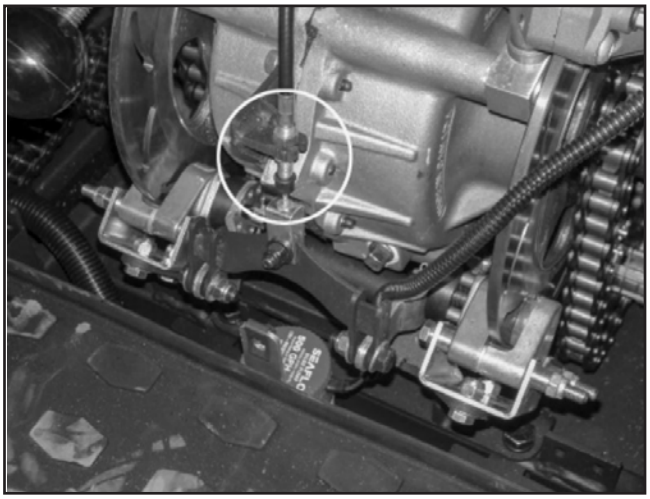
The hand brake lever should be adjusted such that when pulled up firmly it is capable of holding the vehicle from rolling on a grade. It should also ensure a good braking response when applied to stop the vehicle during normal operation.

CAUTION

If the emergency/parking brake system is adjusted too tight when the lever is in the down position, overheating of the brake system will occur due to drag between brake pads and brake discs.

Parking Brake System (Frontier Models)

Adjusting the Emergency/Parking Brake



1. Remove the firewall.
2. Ensure the parking brake lever is fully up.
3. Locate the Parking Brake Adjustment Bracket near the bottom of the transmission. Check to ensure that the brakes are **NOT** engaged when the Brake Lever is in the up & off position. Using a 0.004" feeler gauge slip it between the emergency/parking brake pad and brake disc. Ensure that you push the opposite side pad up against the brake disc before setting this gap
4. If the gap in not within spec, locate the castle nut at the mechanical brake cam lever and remove the cotter pin.
5. Loosen the castle nut until it can be threaded by hand.
6. Using a 0.004" feeler gauge or a piece of regular photo copy paper (such as used for these instructions), slip it between the emergency/parking brake pad and brake disc. Ensure that you push the opposite side pad up against the brake disc before setting this gap.
7. Slowly hand tighten the castle nut until the feeler gauge (or piece of paper), becomes snug between the pad and brake disc.
8. Back off the castle nut just enough for a new cotter pin to be installed. The feeler gauge (or piece of paper), should pull out at this point with just the slightest bit of resistance.
9. Check for drag by driving without activating any brakes for about 100 feet. Stop and check for heat on the brake discs. They should both be cool (or no hotter than the beginning of the test). Adjust if necessary.
10. Check the effectiveness of the emergency brake by activating it while parked on a grade. Re-adjust the brakes if necessary.
11. The emergency/parking brake should be checked for proper adjustment every 25 hours.

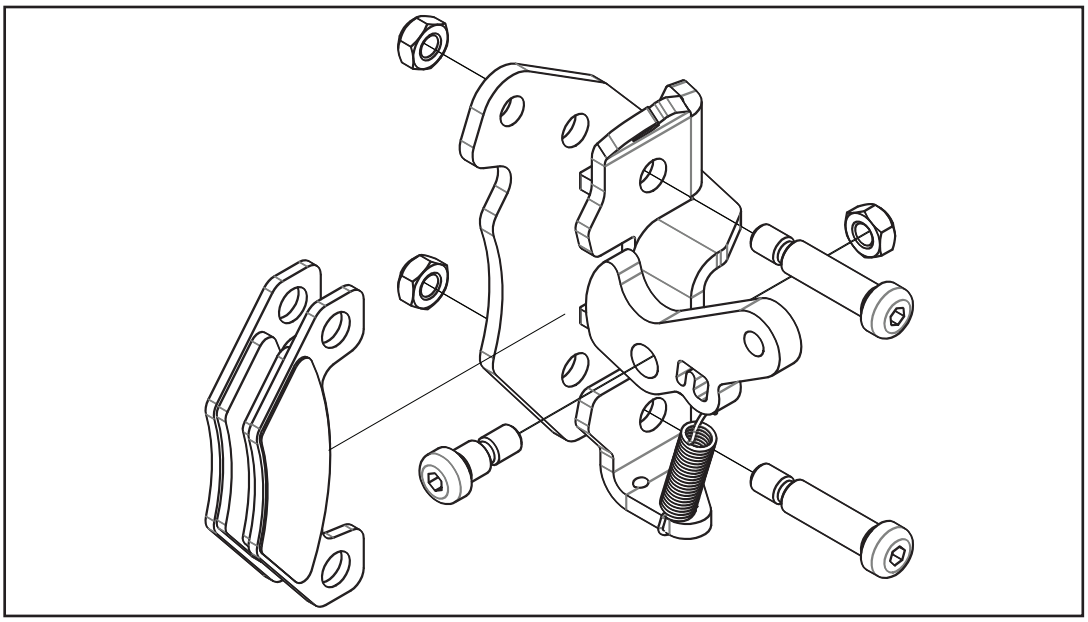
⚠ NOTE

Oil on the brake disc caused by improper chain oiling can permanently reduce the effectiveness of all brake systems.

Removing The Parking Brake Pads (Aurora Models)

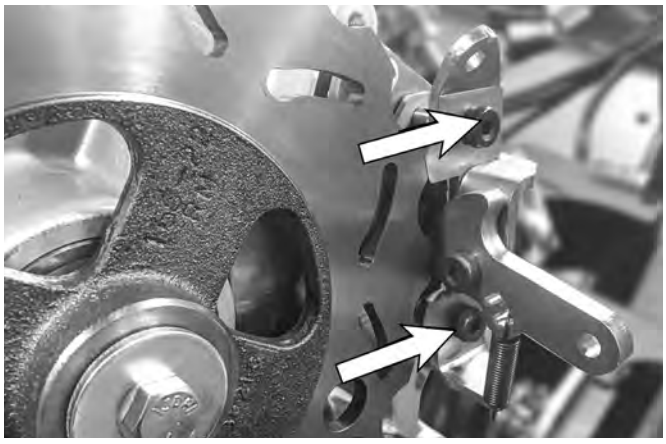
The Parking Brake Disc and Caliper is located on the right hand side of the transmission on Aurora Models.

1. Remove the hood from the vehicle.
2. Remove the firewall from the vehicle.
3. Reaching through the firewall, locate the hardware securing the pads in the caliper.



4. Disconnect the parking brake cable from the lever on the Park Brake Caliper.
5. Using a 1/2" wrench and 3/16" allen head wrench, remove the two (2) Bolt, Shoulder 3/8 x 1.25 and two (2) Nut, Conelock 5/16-18 securing the brakes in the caliper. Slip the pads from the caliper.



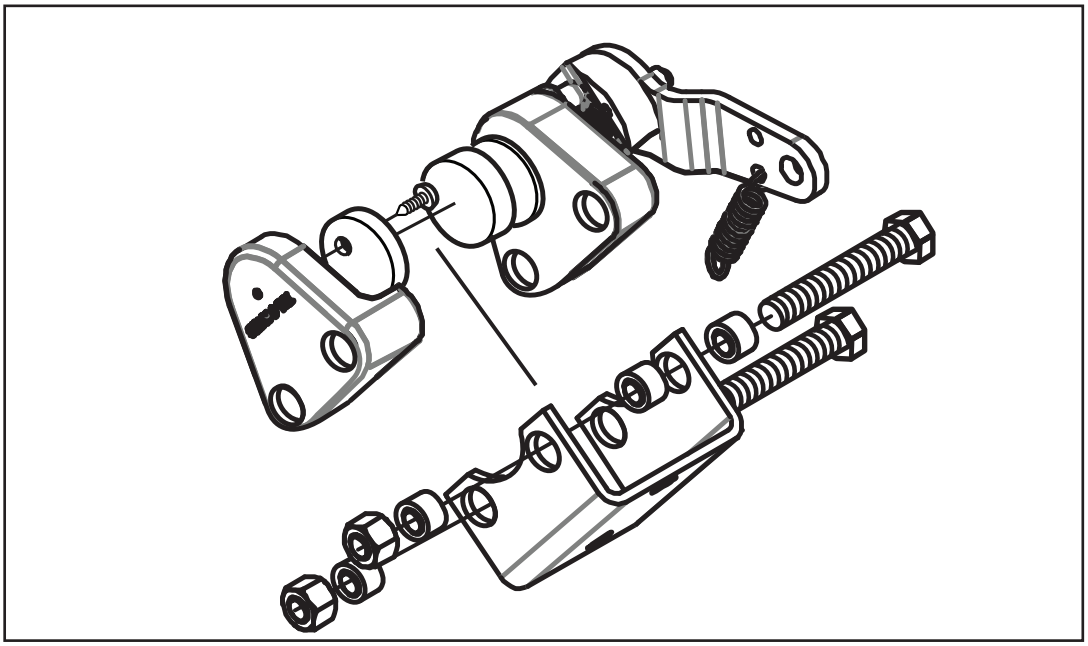


Installing The Parking Brake Pads (Aurora Models)

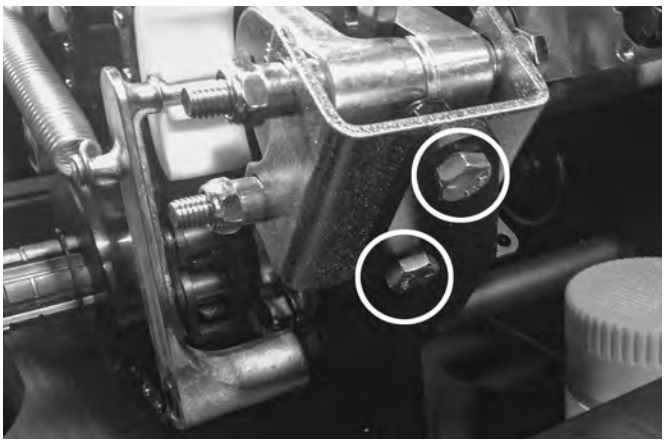
1. Install the brake pads to the parking brake caliper and secure with two (2) Bolt, Shoulder 3/8 x 1.25 and two (2) Nut, Conelock 5/16-18

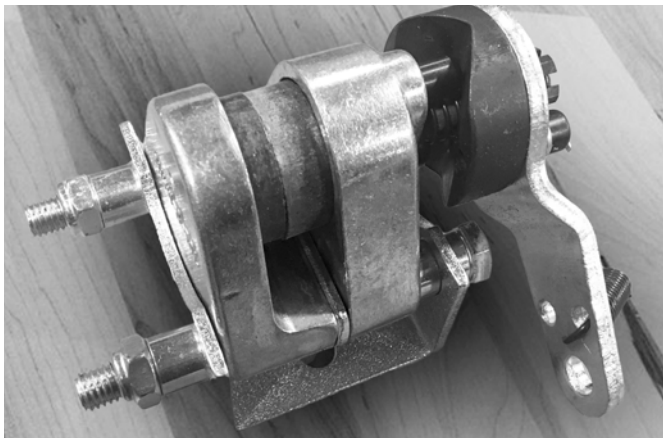
2. Reconnect the parking brake cable to the parking brake arm.
3. To seat the new pads and stretch the cable, apply the parking brake 5-6 times to full on and off.
4. Install the firewall and check park brake efficiency of the vehicle on a grade.

Removing The Parking Brake Pads (Frontier Models)



1. Remove the two (2) HHCS 5/16-18x5/8 Gr5 Zn, two (2) Lockwasher 5/16 Split-Ring Bnz and two (2) Locknut, Serrated Flange 5/16-18 Zn securing the parking brake caliper to the caliper bracket.
2. Remove the caliper.

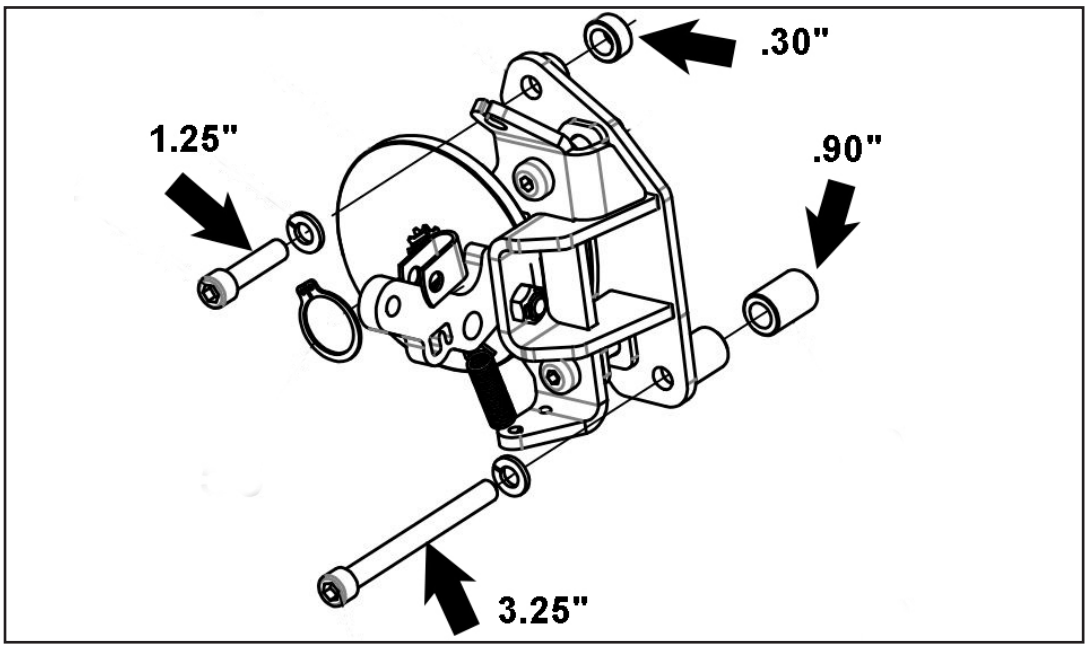




3. Split the caliper assembly. Take note of all spacers, shims and washers within the assembly.
4. Locate the Dead Side Brake Housing and remove the screw securing the brake puck. Remove the brake puck and discard.

5. Locate the Live Side Brake Housing and turn upside down to remove 2nd brake pad from the housing cavity. Please note that there is also a washer placed beneath the brake pad.
6. Lay out both Dead Side and Live Side Brake housings with new brake pads.
7. Install the new brake pad to the "dead side" housing and secure with screw. Apply some Blue LOCTITE 243 to the threads of the screw before installing.
8. Locate the "live" side brake housing and slip the shim washer into the cavity first before installing the brake pad.
9. Install the 'Live" brake pad to the top of the previously installed shim.
10. Reinstall the Parking Brake Caliper to the caliper bracket
11. Loosely snug the Parking Brake Caliper hardware. Squeeze the hand brake a few times to centre the pads and then hold the brake on while tightening the hardware to a final **torque of 209 in.lbs (23.6 Nm).**
12. Perform the servicing procedure, **Adjusting the Emergency/Parking Brake (Frontier Models**

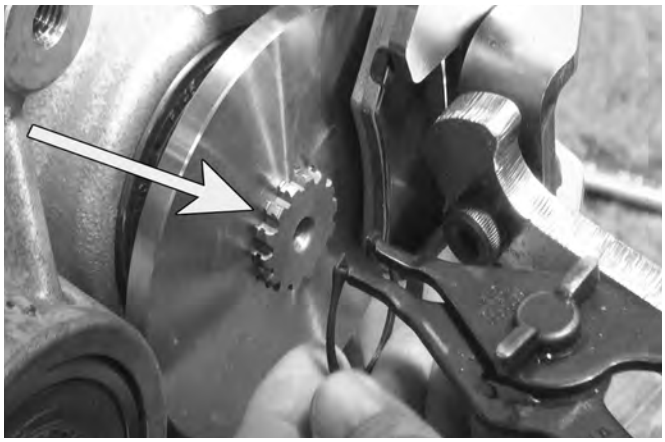
Removing The Parking Brake Disc/Caliper (Aurora Models)

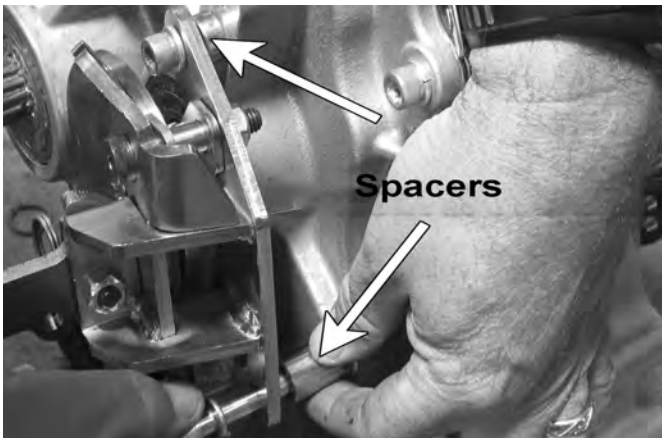


The parking brake disc/caliper is mounted on the right hand side of the transmission.

1. Remove the firewall.
2. Perform the servicing procedure, **Removing The Steering Brake Disc (Aurora Models)**

3. Remove the retaining ring securing the Parking Brake Disc to the shaft.
4. Remove one (1) SHCS 5/16x1.25 ZINC, and one (1) SHCS 5/16-18x3.25 ZINC, securing the Parking Brake Caliper to the transmission housing.
5. Be aware of lock washers and spacers used in the assembly.
6. Slide the parking brake disc and brake caliper off simultaneously.







7. Perform the servicing procedure, **Brake Pad Inspection (Service Brake Pads) Aurora Models Only.**

Installing The Parking Disc Brake/Caliper (Aurora Models)

1. Assemble some new pads to the caliper.
2. Connect the park brake cable to the caliper assembly.

